Total ML Features Specification

MODEL: RANDOM FOREST

AXIOM / Meridian Predictive Maintenance Portal · NASA C-MAPSS FD001

56	14	30	4
TOTAL MODEL FEATURES	DYNAMIC SENSORS	WINDOW CYCLES	STATS PER SENSOR

1. Feature Extraction Methodology

The predictive maintenance regressor models the Remaining Useful Life (RUL) of turbofan engines using strictly causal rolling windows. Observations are extracted across the preceding **30 contiguous operational cycles** ($t-29$ to t). Constant sensors (variance = 0) are eliminated. For each of the remaining 14 sensors, 4 statistical aggregations are computed: **Instantaneous Reading (`_last`)**, **Rolling Mean (`_mean`)**, **Rolling Standard Deviation (`_std`)**, and **Rolling Linear Trend Slope (`_slope`)**.

2. Retained Dynamic Sensors (14 Sensors)

Sensor Key	Component / Parameter Name	Units	Physical Domain
`sensor_2`	LPC outlet temperature	°R (Rankine)	Thermal (Low-Pressure Compressor)
`sensor_3`	HPC outlet temperature	°R (Rankine)	Thermal (High-Pressure Compressor)
`sensor_4`	LPT outlet temperature	°R (Rankine)	Thermal (Low-Pressure Turbine)
`sensor_6`	Bypass duct pressure	psia	Pneumatic / Bypass Airflow
`sensor_7`	HPC outlet total pressure	psia	Pneumatic (High-Pressure Compressor)
`sensor_8`	Physical fan speed	rpm	Kinematic / Rotational
`sensor_9`	Physical core speed	rpm	Kinematic / Core Spool
`sensor_11`	HPC outlet static pressure	psia	Pneumatic (High-Pressure Compressor)
`sensor_12`	Fuel flow ratio to static pressure (Ps30)	pps/psi	Combustion & Fuel Metering
`sensor_13`	Corrected fan speed	rpm	Kinematic (Aerodynamically Corrected)
`sensor_14`	Corrected core speed	rpm	Kinematic (Aerodynamically Corrected)
`sensor_15`	Bypass ratio	ratio	Aero-Thermodynamic Cycle Balance
`sensor_17`	Bleed enthalpy	dataset units	Secondary Air System Thermodynamics
`sensor_20`	HPT coolant bleed	lbm/s	Cooling System Mass Flow
`sensor_21`	LPT coolant bleed	lbm/s	Cooling System Mass Flow

Note: 7 sensors were excluded due to zero operational variance: `sensor_1`, `sensor_5`, `sensor_10`, `sensor_16`, `sensor_18`, `sensor_19`, and operating settings.

3. Complete Catalogue of All 56 Model Features

1. Instantaneous Reading (_last) 14 features

Current observation at cycle t

`sensor_2_last` `sensor_3_last` `sensor_4_last`
`sensor_6_last` `sensor_7_last` `sensor_8_last`
`sensor_9_last` `sensor_11_last` `sensor_12_last`
`sensor_13_last` `sensor_14_last` `sensor_15_last`
`sensor_17_last` `sensor_20_last` `sensor_21_last`

2. Rolling Mean (_mean) 14 features

30-cycle rolling average ($t-29$ to t)

`sensor_2_mean` `sensor_3_mean` `sensor_4_mean`
`sensor_6_mean` `sensor_7_mean` `sensor_8_mean`
`sensor_9_mean` `sensor_11_mean` `sensor_12_mean`
`sensor_13_mean` `sensor_14_mean` `sensor_15_mean`
`sensor_17_mean` `sensor_20_mean` `sensor_21_mean`

3. Rolling Standard Deviation (_std) 14 features

30-cycle rolling volatility / dispersion ($ddof=0$)

`sensor_2_std` `sensor_3_std` `sensor_4_std`
`sensor_6_std` `sensor_7_std` `sensor_8_std`
`sensor_9_std` `sensor_11_std` `sensor_12_std`
`sensor_13_std` `sensor_14_std` `sensor_15_std`
`sensor_17_std` `sensor_20_std` `sensor_21_std`

4. Rolling Linear Trend (_slope) 14 features

30-cycle regression slope (dy/dt degradation rate)

`sensor_2_slope` `sensor_3_slope` `sensor_4_slope`
`sensor_6_slope` `sensor_7_slope` `sensor_8_slope`
`sensor_9_slope` `sensor_11_slope` `sensor_12_slope`
`sensor_13_slope` `sensor_14_slope`
`sensor_15_slope` `sensor_17_slope`
`sensor_20_slope` `sensor_21_slope`

4. Top 10 Most Influential Features (Random Forest Importance)

Rank	Feature Identifier	Physical Significance	Gini Importance
#1	**sensor_17_mean**	Bleed Enthalpy rolling average	**26.04%**
#2	**sensor_3_mean**	HPC Outlet Temperature rolling average	**21.39%**
#3	**sensor_11_slope**	HPC Static Pressure degradation slope	**12.69%**
#4	**sensor_9_slope**	Physical Core Speed rate of change	**3.59%**
#5	**sensor_11_mean**	HPC Static Pressure rolling average	**3.22%**
#6	**sensor_12_slope**	Fuel Flow / Ps30 trajectory slope	**2.95%**
#7	**sensor_7_mean**	HPC Total Pressure rolling average	**1.85%**
#8	**sensor_4_slope**	LPT Outlet Temperature trajectory slope	**1.85%**
#9	**sensor_2_mean**	LPC Outlet Temperature rolling average	**1.71%**
#10	**sensor_9_mean**	Physical Core Speed rolling average	**1.48%**